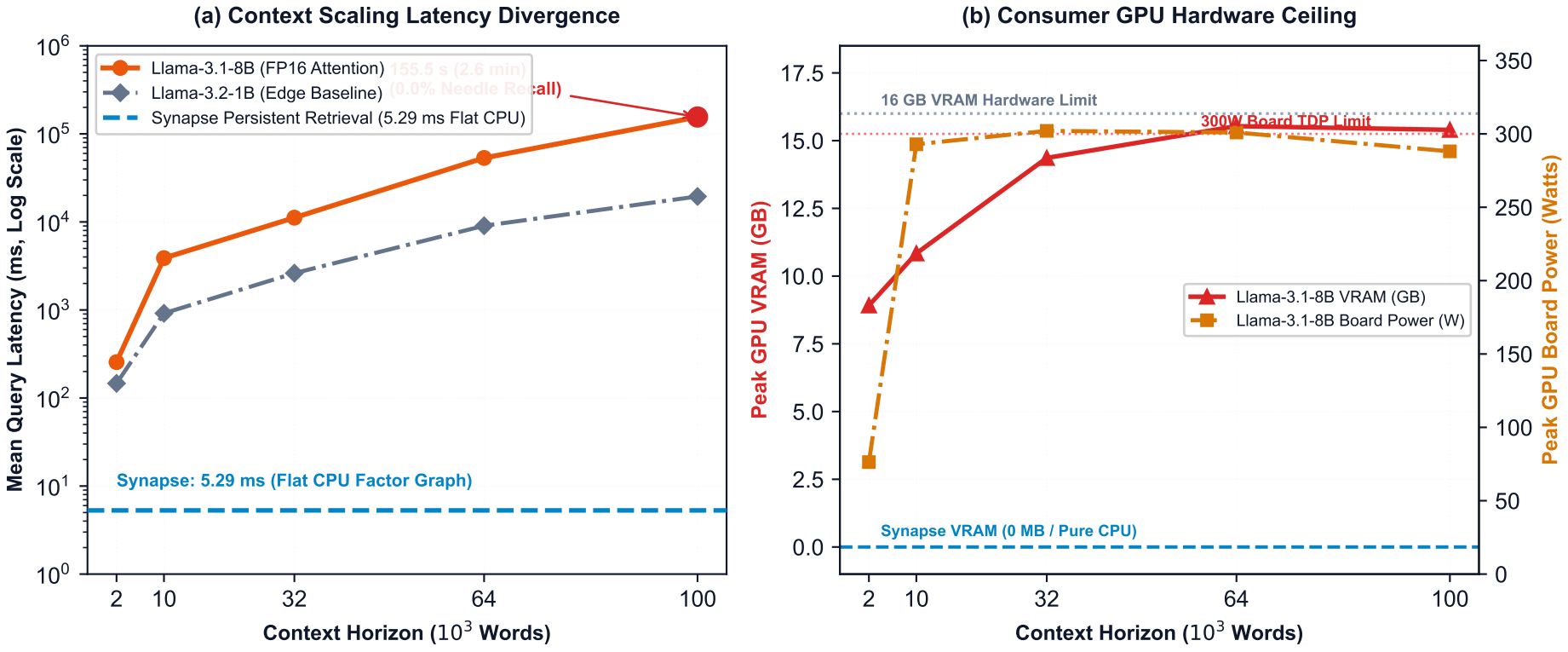
(a) Context Scaling Latency Divergence
Llama-3.1-8B (FP16 Attention)
Llama-3.2-1B (Edge Baseline)
Synapse Persistent Retrieval (5.29 ms Flat CPU)
Synapse: 5.29 ms (Flat CPU Factor Graph)
Mean Query Latency (ms, Log Scale)
Context Horizon (10^3 Words)
(b) Consumer GPU Hardware Ceiling
16 GB VRAM Hardware Limit
300W Board TDP Limit
Llama-3.1-8B VRAM (GB)
Llama-3.1-8B Board Power (W)
Synapse VRAM (0 MB / Pure CPU)
Peak GPU VRAM (GB)
Peak GPU Board Power (Watts)
Context Horizon (10^3 Words)